

Athyam Ramcharan

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Summary

AI & Robotics Engineering student specializing in ROS 2-based autonomous systems, machine learning, and computer vision. Hands-on experience architecting full robotics stacks — from ROS 2 node design, SLAM, and sensor fusion in Gazebo simulation to embedded hardware deployment on Raspberry Pi and FPGA. Competing in national-level competitions (Smart India Hackathon, e-Yantra IIT Bombay). Active researcher in low-resource LLMs and quantum ML. Seeking roles in robotics software, autonomous systems, or AI/ML engineering.

Education

Dayananda Sagar University

B.Tech. in Artificial Intelligence and Robotics Engineering

Bangalore, India

Aug 2024 – Present

Sir M Visvesvaraya Polytechnic

Diploma in Mechanical Engineering

Hospet, India

Nov 2021 – 2024

Projects & Research

Low-Resource LLM & Edge AI | Qwen, Gemma, PyTorch, HuggingFace

Ongoing

- Fine-tuned Qwen 0.5B/7B for Hindi and Kannada using SFT/QLoRA; built a Qwen 72B teacher-student pipeline producing 8,366 instruction pairs.
- Studied model-capacity thresholds: 0.5B showed syntactic/news mimicry, while 7B retained semantic instruction-following on the same noisy data.
- Deployed Gemma 3 1B on a 4GB Jetson Nano using 8-bit quantization and LoRA; evaluated memory, latency, and compression trade-offs.

Quantum-Enhanced ML for Disease Detection | Qiskit, PennyLane, Scikit-learn

Ongoing · Smart India Hackathon

- Building a hybrid quantum-classical ML pipeline for early disease prediction using variational quantum circuits and classical neural networks.
- Benchmarking quantum vs. classical classification accuracy on medical datasets using IBM Quantum and Azure Quantum.

Autonomous Agricultural Drone System | ROS 2, Nav2, Gazebo, SLAM Toolbox, OpenCV

Stage I · e-Yantra IIT Bombay

- Architected a multi-node ROS 2 system for autonomous drone control with modular perception, planning, and actuation packages.
- Built LIDAR/IMU/camera sensor fusion and deployed SLAM Toolbox + Nav2 for pose estimation, mapping, path planning, and obstacle avoidance.
- Preparing Swift Pico Rev2 hardware bring-up with ros2_control, real-time controller tuning, and sim-to-real validation.

Technical Skills

Languages: Python, C++, C, MATLAB, Bash/Linux

ML/AI Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, OpenCV, NLTK

Robotics & Embedded: ROS 2 (Nav2, tf2, ros2_control, rclpy/rclcpp), Gazebo, SLAM Toolbox, Sensor Fusion, MoveIt 2, Raspberry Pi, Arduino, LIDAR, IMU, FPGA

Quantum Computing: Qiskit, PennyLane, hybrid classical-quantum algorithms, IBM Quantum / Azure Quantum

Cloud & DevOps: AWS (Cloud Practitioner), Google Cloud, Azure AI Foundry, Docker, Git/GitHub, Linux (Ubuntu, Arch)

Tools: Jupyter, Kaggle, SOLIDWORKS, ANSYS, CNC (G-code/M-code), RapidMiner

Certifications

RapidMiner Certified Professional – Machine Learning

Oct 2024

AWS Cloud Practitioner Essentials – Amazon Web Services

AWS Technical Essentials – Amazon Web Services

Industrial Exposure

Moog India – Aerospace, Robotics & Precision Control Systems: Gained exposure to flight control hardware, aerospace-grade actuators, and closed-loop control systems.

GTTC – Industrial Robotics, CNC Machining & Control Systems: Hands-on training in CNC programming (G-code/M-code) and industrial automation.

Difacto Robotics & Automation – Industrial Robots (Fanuc, ABB, Kawasaki): Programmed and operated industrial robotic arms for manufacturing automation workflows.

Achievements & Competitions

Smart India Hackathon 2024 | National-level competition, Quantum ML track

e-Yantra Robotics Competition, IIT Bombay | Autonomous Drone – Stage I qualifier